

Sujit Daniel

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EDUCATION

University of Arizona

Tucson, AZ

B.S. in Computer Science, Minors: Mathematics & Finance

Aug. 2022 – May 2026

- **Awards:** 50% tuition merit scholarship; Dean's List (4× total); Academic Distinction
- **Relevant Courses:** Compilers & Systems Software, Systems Programming, Machine Learning, Stochastic Calculus

EXPERIENCE

Machine Learning Engineer | *Python, PyTorch, R, Linux, HPC, Git*

Aug. 2026 – Present

University of Arizona College of Medicine

Tucson, AZ

- Achieved **0.883** Dice against expert labels by training deep learning segmentation models on a **250+** subject cohort
- Run training and evaluation on **HPC** infrastructure, benchmarking every model against expert-labeled ground truth
- Build preprocessing and quality control pipeline for **3D** image volumes and paired tabular data
- Maintain reproducible Python and R pipelines that rerun full analysis end to end from raw inputs

Artificial Intelligence Intern | *FastAPI, React, TypeScript, OpenAI API, n8n, Docker*

Jun. 2025 – Aug. 2025

Genmark AI

Seattle, WA

- Served comment suggestions in under **2s** by building a FastAPI backend for profile analysis and post classification
- Built Chrome extension surfacing AI-generated comment suggestions on LinkedIn profiles on demand
- Automated LinkedIn data capture into the LLM pipeline with n8n workflows

Software Engineer Intern | *React, Next.js, GraphQL, TypeScript, Vercel*

May 2025 – Aug. 2025

Lynnapse (golynnapse.com)

Remote

- Built production React/Next.js UI from reusable atoms and molecules in a **1,000+** component design system
- Integrated GraphQL queries surfacing research matches, replacing manual cold outreach to professors
- Defined API contracts and reviewed PRs to keep frontend and backend integrations aligned

Undergraduate Research Assistant | *C++, Python, OpenCV, NumPy, Raspberry Pi, Linux*

Nov. 2023 – May 2025

Steward Observatory | NASA-Funded

Tucson, AZ

- Achieved sub-degree RMS pointing error by fitting a transform over **27** reference points from **154** features per image
- Built C++/Python vision pipeline mapping camera images to a calibrated transform for telescope pointing
- Versioned pipeline in Git for single-command reproducibility; presented results to lab and funders

RESEARCH & PUBLICATIONS

S. Daniel et al. "Deep Segmentation of White Matter Hyperintensities Without Manual Tracing." *In preparation*, 2026.

PROJECTS

NetOpt: MLIR Compiler for Packet Processing | *C++17, MLIR, TableGen, CMake*

Mar. 2026 – May 2026

- Built MLIR compiler framework for packet processing, modeling how a network card spreads work across CPU cores
- Implemented lock-free multi-consumer queue in C++17; stress-tested **1–8** consumers, benchmarked **15.6M+** items/s
- Built compiler passes that eliminate unused queues and parallelize affine loops free of state conflicts
- Verified both passes with **16** FileCheck tests covering safe and unsafe transformation cases

Earnings Intelligence Agent | *Python, Claude API, FastAPI, pgvector, Voyage AI, Docker*

May 2026 – Jun. 2026

- Built SEC-filings retrieval pipeline ingesting 10-Qs via EDGAR into pgvector using voyage-finance-2 embeddings
- Built Claude agent retrieving top-k filing evidence via cosine similarity to compare companies and cite sources
- Evaluated retrieval on **18** questions at **90%** Precision@5; reduced input cost **37%** through context optimization

Stock Direction & Volatility Targeting | *Python, scikit-learn, pandas, yfinance, matplotlib*

Jan. 2025 – Jun. 2025

- Ran **126** k-NN/SVM/RF experiments with chronological splits and train-only scaling; **50.34%** mean accuracy
- Found no directional edge; added volatility targeting, cutting max drawdown from **-34.5%** to **-16.8%**

SKILLS

Programming Languages: Python, C, C++, R, MATLAB, SQL, TypeScript, JavaScript, Java

ML & Data: PyTorch, scikit-learn, pandas, NumPy, OpenCV, embeddings, RAG, pgvector, PostgreSQL, MongoDB

Backend & Web: FastAPI, REST APIs, React, Next.js, GraphQL, Claude API, OpenAI API

Infrastructure & Tools: Linux/Unix, HPC, MLIR, TableGen, CMake, Docker, AWS (S3, Lambda), Git, CI/CD, PyTest